VOIDSCOPE · DESIGN DECISION RECORD

VoidScope steerable camera tip (R11)

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	VDS-DDR-003	Version	0.1
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/voidscope		
Source	docs/decisions/0003-steerable-camera-tip.md @ 463eaf8		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Amish's requirement decision 3A carried out: cable-steered camera tip, R11 restated, the changes it needed and their results

0003: Steerable camera tip (R11)

- **Date:** 2026-10-03
- **Status:** accepted. Decided by Amish on 2026-10-03, who chose option A on every requirement decision put to him: "1A 2A 3A 4A 5A 6A 7A 8A 9A 10A 11A". Decision 3A is VoidScope's.

Context

At TRL 3 the constructable design (VDS-DDR-002) met every requirement but R11, "pass a 4 m test channel with two 45° bends". The rigid rod goes straight only, and a flexible lead section had been left to a later version (VDS-DDR-001 D12). Option A, which Amish chose, keeps the rigid rod and instead lets the camera look around a bend: a cable-steered articulating camera tip that bends up to 90° each way, controlled from the handle, with R11 restated as "see around a 45-degree bend from a straight hole" and given a verification method. The rigid rod, the air path through its bore and the water tube to the bite valve stay as they are, and the tip must still pass the 51 mm hole.

Options considered

The options were put to Amish with the requirement decisions; he chose A. The design choices below were made to carry A out.

Table 1. How the tip is built: choices made.

Question	Options looked at	Chosen	Why
Bending section	Notched single tube (laser cut); rolling-contact links; pinned hinge links	Five pinned joints: a turned base link, four printed links and a turned tip housing, joined by 2 mm stainless pins	Pinned links can be printed by any SLS service and built with hand tools; a notched tube needs laser cutting and fatigues
Bending planes	One plane (two wires); two planes (four wires)	One plane, sideways (left and right of the bite valve)	R11 needs one plane; turning the rod picks the plane; two wires fit beside the existing lines, four would not
Where the camera goes	Keep it in the nose; move it to the tip	In a turned tip housing on the last joint, its window 3 mm behind the front edge	A camera behind the bend cannot see around it
How the wires reach the handle	Separate cable outside the rod; inside the rod bore	Two 3 mm PTFE-lined steel-coil housings inside the rod, stopping in the collar's front wall at the head end and in the control body at the handle	No loose cable outside; the housings stay threaded with the other lines (tent-pole storage)
Handle control	Bought borescope handle; drum and lever	A printed clamp body on section 4 ahead of the grip, with an 18 mm drum on a shaft, a thumb lever and a friction lock	Built from stock and printed parts; the lever sits where the rear hand's thumb reaches
Sealing the joints	Bare links; a sheath	A 0.6 mm silicone sheath over the bending section, replaceable, with a spare in the kit	Keeps grit out of the pins; the camera itself is already IP68

Decision

Table 2. Changes made to the model, the BOM and the calculations.

#	Before	Now	Why
T1	Camera in the nose against a 26 mm lip, clamped by a rubber spacer ring	Camera in a tip housing (33 mm outside, 29.3 mm bore, 66 mm long) held by two nylon-tipped M3 grub screws; the spacer ring is gone	The camera must be ahead of the bend
T2	Nose bore 29.3 mm from 12 to 72 mm, then the lip	Nose bore 29.3 mm right through; two M3 tapped holes near the front for the base link's grub screws	Takes the base link's spigot
T3	Nothing ahead of the nose	Base link (turned, 29.25 mm spigot 14 mm into the nose, 33 mm flange), four printed PA12 links (33 mm outside, 23 mm bore, 10 mm bodies, 5.25 mm gaps), tip housing; ten 2 mm pins; nested lugs top and bottom with their ends rounded about the pins	Each joint closes at 18.1°, so five give 90° each way; the rounded lug ends keep neighbouring links clear at full bend (checked in the model at 90° each way)
T4	No steering	Two 0.6 mm 7x7 stainless wires, 14 mm either side of the camera axis, from crimped anchors in the tip housing, through every link and a slanted hole in the base link, along the nose bore and through the collar's front wall into their housings	Pulling one wire bends the tip toward it
T5	No housings in the rod	Two 3 mm housings in the rod bore 10.5 mm either side and 7.5 mm below the axis, clear of the snap button springs (2.0 mm), the coupler bores (0.95 mm), the cable (6.0 mm) and the guide tube (5.0 mm)	The only place in the bore below the spring strips that clears everything
T6	Collar outlets at 0, 45, 135, 225, 270 and 315°	Lower side outlets moved to 240 and 300°; two 3.2 mm housing stops 2 mm deep in the plenum side of the front wall, 1.2 mm wire holes on through the spigot	The housings pass the outlets where 225 and 315° would have put the holes on top of them; six outlets of the same size, so the air path is unchanged
T7	Section 4 plain behind the coupler	Two 4 mm holes on top of section 4, 332 and 368 mm from its back end	Where the housings leave the rod for the control
T8	Nothing on the rod ahead of the grip	Steering control: printed PA12 clamp body 8 mm ahead of the grip, 18 mm aluminium drum on a 6 mm shaft between two cheeks, 50 mm thumb lever, friction lock knob, M4 clamp screw	71° of lever gives 90° at the tip; about 5 N at the thumb holds it there
T9	Bite valve ending 2 mm behind the nose front	Unchanged position, 2 mm behind the nose front; the tip now extends 140 mm beyond it	The water line is kept as Amish asked; access to the valve is a new open decision (VDS-DEC-001)
T10	Lathe allowance 3 h (USD 180)	4 h (USD 240) for the base link, tip housing and drum	Three more turned parts

R11 is restated in VDS-REQ-001 v0.4: "see around a 45-degree bend from a straight hole", verified in a test box with a 51 mm hole into a 300 mm wide passage that turns 45° 0.5 m beyond the hole, reading 20 mm letters 1 m down the turned passage, to each side, then straightening the tip and withdrawing it.

Consequences

Table 3. Results (VDS-CAL-001 v0.2).

Requirement	Before	Now
R11, see around a 45° bend (restated)	Not met (rigid rod, old wording)	Met on paper: 90° each way available, 45° needed; bent 45° the tip needs 83 mm to the side; 19 pixels across a 20 mm letter at 1 m
R1, 51 mm core hole	48 mm	48 mm with the tip locked straight; every tip part inside the outline in the model
R2, reach	4,063 mm from the grip	4,131 mm can go in before the steering control reaches the hole (4,203 mm from the grip)
R9, setup	2.1 min	2.2 min (0.1 min to check the steering)
R8, cost	USD 1,578	USD 1,780: Value-engineering target: USD 2,000. Estimated cost of the constructable design: USD 1,780 (USD 220 under the target)
Probe mass	3.66 kg	4.18 kg (tip 0.28 kg ahead of the nose; housings, wires and control 0.41 kg)
Rod buckling	1,001 N, 5.0 times a 200 N push	940 N, 4.7 times (longer probe, heavier line)
Handling, 3 m unsupported	34 N at the grip, 2.55 on yield at a button hole	32 N at the grip, 2.33 on yield

The model now runs 180 constructability checks, all passing. The new parts to watch at TRL 4 are the printed lugs (8.7 MPa bearing at a 30 N thumb push against the stop, about a fifth of PA12's strength), the 0.6 mm sheath against sharp rubble, the wire's breaking load (a factor of 3.0 assumed) and the camera cable's real bending stiffness, which sets the thumb force. The wires are not self-locking, so a bent tip straightens against the hole's edge if it is pulled back; the procedure is still to steer straight and lock before going in or out.

New for Amish: with the tip 140 mm ahead of the bite valve, the survivor reaches the valve past the camera tip. Options and a recommendation are in the design decisions register (VDS-DEC-001, open decision 1).